

Retrieval-Augmented Generation (RAG) - MNLP HW2

Antonio Serra 1893530 - Daniele Falanga 1918515

1 Introduction

This report describes our Homework 2 system for augmented generation and evaluation. Starting from the semantic retrieval setting of Homework 1, we implemented a complete Retrieval-Augmented Generation (RAG) pipeline in which retrieved evidence passages are used to condition small open-weight language models. The goal is to generate short factual answers to natural-language questions and compare three generation settings: a **baseline** relying only on parametric knowledge, a **RAG** setting using retrieved chunks, and an **oracle** setting where the gold chunk is always injected as first evidence.

Our experiments focus on two small language models, Qwen3-0.6B and SmoLLM2-360M-Instruct, both below the 3B parameter constraint. We evaluate the systems on the test split using automatic metrics, LLM-as-a-Judge evaluation on a 200-sample subset, and manual validation by both group members. As additional requirement, we also explore Wikidata-based context enrichment.

2 Methodology

For each test example, the input consists of a question, candidate chunks, and a target short answer. We used a fine-tuned all-MiniLM-L6-v2 retriever from Homework 1 to rank the candidate chunks and selected the top- k passages, with $k = 3$, as required. In the baseline setting, the model receives only the question. In the RAG setting, the prompt contains the top-3 retrieved chunks followed by the question. In the oracle setting, the correct chunk is always placed in first position; if it is already retrieved, it is moved to the top, otherwise it replaces the lowest-ranked retrieved chunk.

Prompts were designed to encourage concise factual answers. The models were used in zero-shot mode, without examples in the prompt. Generated answers were post-processed by removing possible

answer prefixes and keeping the first non-empty line, in order to make the output closer to the expected short-answer format.

For the additional requirement, we implemented a Wikidata-enriched RAG variant. For each entity, we queried Wikidata using the provided `wikidata_id` and cached English labels and descriptions. The description was injected as additional knowledge before the retrieved context. This was intended to act as a semantic anchor, especially when retrieved chunks contained multiple entities or partially noisy evidence.

3 Experimental Setup

The dataset is the MNLP 2026 Homework dataset, using the 2,000-example test split for evaluation and the blind split for final output generation. The two evaluated models are Qwen3-0.6B and SmoLLM2-360M-Instruct. For both models, we generated outputs in baseline, RAG, and oracle configurations on the full test set. For the blind split, we produced the required RAG output files.

Automatic evaluation was performed using Exact Match (EM), substring Exact Match (subEM), and METEOR. EM is strict and rewards only exact string matches with the gold short answer. subEM is more permissive and checks whether the gold short answer appears inside the generated answer. METEOR provides a softer similarity measure based on lexical overlap and related matching mechanisms.

For LLM-as-a-Judge, we selected the best RAG system according to METEOR, namely SmoLLM2-360M-Instruct in the RAG setting. We sampled 200 test examples and asked an LLM judge to assign a binary score: 1 if the correct short answer was present in any valid form, and 0 otherwise. The same 200 examples were manually annotated by both group members using the same criterion. We then computed Cohen's Kappa between the two

human annotators and between each annotator and the LLM judge.

4 Results

Table 1 reports the main automatic results. Both models benefit strongly from retrieved context. The baseline scores are very low, showing that the small models have limited parametric knowledge for this task. RAG substantially improves all metrics for both models, while oracle gives the best results, confirming that retrieval quality is a major bottleneck.

Model	Setup	EM	subEM	METEOR
Qwen3	Baseline	0.30	2.15	5.41
Qwen3	RAG	5.65	29.75	30.63
Qwen3	Oracle	6.70	33.75	34.33
SmolLM2	Baseline	0.75	6.15	8.96
SmolLM2	RAG	15.85	32.25	32.88
SmolLM2	Oracle	16.90	34.05	34.62

Table 1: Main generation results on the test split.

SmolLM2-360M-Instruct is the best RAG model, reaching 15.85 EM, 32.25 subEM, and 32.88 METEOR. Interestingly, despite being smaller than Qwen3-0.6B, it performs better in all configurations. This suggests that instruction tuning and output format alignment are more important than parameter count in this specific short-answer RAG setting.

On the 200-sample manually inspected subset, the best RAG system obtained 14.50 EM, 32.00 subEM, and 35.29 METEOR. The LLM judge assigned a positive score to 96.50% of the samples. However, agreement analysis revealed a strong mismatch between the LLM judge and human annotators: Cohen’s Kappa was 0.9898 between the two humans, but only 0.0672 and 0.0688 between each human and the LLM judge. This indicates that the judge was too permissive and often accepted answers that were semantically related but factually wrong. For example, it sometimes accepted wrong entities when the generated answer was plausible or partially overlapping with the question context.

The qualitative analysis highlights three main failure modes. First, noisy or irrelevant retrieved chunks can mislead the model, especially when they contain entities of the same type as the correct answer. Second, the models show sensitivity to context ordering: earlier chunks often receive more attention, which explains why the oracle setting improves performance when the correct evidence is placed first. Third, some questions are ambiguous or require very specific entity disambiguation,

making exact short-answer generation difficult.

The Wikidata-enriched RAG experiment produced a small but positive effect on softer metrics. Compared with standard RAG for SmolLM2-360M, Wikidata enrichment changed EM from 15.85 to 15.75, subEM from 32.25 to 32.70, and METEOR from 32.88 to 33.06. This suggests that Wikidata descriptions can help the model capture the relevant entity more often, although the additional text may also make some answers slightly more verbose, hurting strict EM.

Overall, the results confirm the usefulness of RAG for small language models. The gap between RAG and oracle shows that better retrieval remains the most promising direction for improvement. At the same time, the poor agreement between LLM judge and humans shows that automatic judging should be manually validated, especially in short-answer factual QA where small entity mistakes completely change correctness.

5 Appendix

The following page reports the complete visual comparison of the automatic evaluation results. The figure summarizes EM, subEM, and METEOR scores for both models across the three evaluated configurations: baseline, RAG, and oracle. This chart is useful to visually inspect the improvement obtained by adding retrieved context and the remaining gap between the RAG and oracle settings.

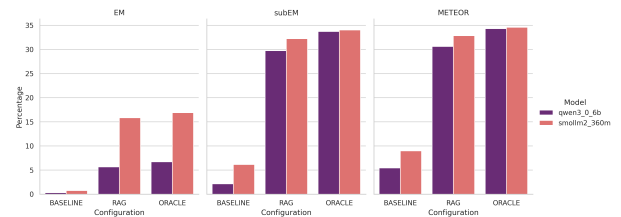


Figure 1: Performance comparison across Baseline, RAG, and Oracle configurations for both models using EM, subEM, and METEOR.